Marking guide

Year 10 Mathematics Yearly Test 2026 · 55 marks · 1 hour 15 minutes. Answers for every question are below, and the test pages that follow have the answers written in red.

Q	Topic and where it comes from	Answer	Marks	Marking notes
Section A · Financial mathematics · 15 marks · Chapter 1 review and booklet				
1	Percentage of an amount Ch 1 review MC 1 · Booklet 1.1	**B ($24.92)**	1	
2	GST Ch 1 review MC 3 · Booklet 1.2	**A ($5.00)**	1	
3	Discount Ch 1 review MC 7 · Booklet 1.3	**D ($41.44)**	1	
4	Simple interest Ch 1 review MC 9 · Booklet 1.6	**C ($120.00)**	1	
5	Percentage increase Ch 1 review SA 2 · Booklet 1.2	**$1 084.16**	1	
6	Mark-up and profit Ch 1 review SA 4 · Booklet 1.3	**(a) $11.40** **(b) $3.80**	2	(b) Follow-through from (a): accept selling price − 7.60.
7	Wages and overtime Ch 1 review MC 4 (a) · Booklet 1.4 (b)	**(a) $473.40** **(b) $94.68**	2	
8	Commission Ch 1 review SA 12 · Booklet 1.5	**$35.55**	1	
9	Gross and net pay Ch 1 review SA 10 · Booklet 1.5	**(a) $1 040.04** **(b) $874.95**	2	(b) Follow-through: accept their (a) − 165.09.
10	Loans Ch 1 review MC 8, ER 2 · Booklet 1.6	**$240.00**	1	Total repaid 60 × 24 = $1440; interest 1440 − 1200 = $240.
11	Compound interest Ch 1 review ER 1 · Booklet 1.7	**(a) $5 624.32** **(b) $624.32**	2	(b) Follow-through: accept their (a) − 5000. Accept $5624.32 rounded to $5624.
Section B · Measurement · 15 marks · Chapter 2 review and booklet				
12	Units of length Ch 2 review MC 1 · Booklet 2.1	**C (1 000)**	1	
13	Area of a rectangle Ch 2 review MC 5 · Booklet 2.4	**D (12 cm^2)**	1	
14	Volume of a cube Ch 2 review MC 9 · Booklet 2.7	**B (27 cm^3)**	1	
15	Scientific notation Ch 2 review SA 8a	**A (2.32×10^5)**	1	
16	Converting units Ch 2 review SA 1a (a) · Booklet 2.1 (b)	**(a) 5 300 m** **(b) 2 500 mL**	2	
17	Perimeter Ch 2 review SA 3 · Booklet 2.2	**38 m**	1	
18	Circumference and area of a circle Ch 2 review SA 4 · Booklets 2.3, 2.5	**(a) 18.8 m** **(b) 28.3 m^2**	2	Also accept 18.85 and 28.27 (2 d.p.), or 18.84 and 28.26 ($\pi \approx 3.14$).
19	Area of a triangle and a trapezium Ch 2 review MC 6, SA 6 · Booklet 2.4	**(a) 30 cm^2** **(b) 20 cm^2**	2	
20	Surface area of a prism Ch 2 review SA 7b · Booklet 2.6	**40 cm^2, 16 cm^2; surface area 136 cm^2**	2	1 mark: both rows correct (20, 40 and 8, 16). 1 mark: total 136 cm^2.
21	Volume and capacity of a cylinder Ch 2 review MC 10, SA 9 · Booklet 2.7	**(a) 75.4 cm^3** **(b) 75.4 mL**	2	Also accept 75.40 or 75. (b) Follow-through: same number as (a), in mL.

Q	Topic and where it comes from	Answer	Marks	Marking notes
Section C · Statistics and probability · 14 marks · Chapter 4 review and booklet				
22	Types of data Ch 4 review MC 6 · Booklet 4.1	**D (categorical and nominal)**	1	
23	Reading a stem-and-leaf plot Ch 4 review MC 7 · Booklet 4.5	**C (24)**	1	
24	Mode Ch 4 review MC 8 · Booklets 4.4, 4.5	**B (31)**	1	
25	Probability Ch 4 review SA 1b	**C ($\frac{1}{2}$)**	1	
26	Mean, median and range Ch 4 review MC 9, SA 8 · Booklet 4.4	**(a) 6** **(b) 5** **(c) 12**	3	Order: 1, 3, 3, 5, 8, 9, 13.
27	Probability Ch 4 review SA 2	**(a) 2/8 = 1/4** **(b) 3/8**	2	
28	Time-series graphs Booklet 4.7	**upward trend**	1	
29	Frequency tables, histograms and shape Ch 4 review SA 6 · Booklets 4.2, 4.3, 4.6	**(a) 20** **(b) Columns of height 3 and 2, touching, shown on the graph.** **(c) 45%** **(d) skewed**	4	(a) 20 (b) columns of height 3 and 2, touching (c) 9 ÷ 20 × 100 = 45% (d) skewed. (c) Follow-through from their total.
Section D · Other Year 10 topics · 11 marks · Rest of the Year 10 syllabus				
30	Index laws Syllabus: Indices	**A (a^7)**	1	
31	Substitution Syllabus: Algebraic techniques	**18**	1	
32	Solving equations Syllabus: Equations	**(a) 5** **(b) 10**	2	Accept any correct method, or the answer checked by substitution.
33	Table of values and graphing a line Syllabus: Linear relationships	**(a) y = 1, 3, 5, 7** **(b) points plotted and joined** **(c) gradient 2, y-intercept 1**	3	(a) 1 mark: at least three values correct. (b) Follow-through from their table. (c) 1 mark: both correct.
34	Pythagoras' theorem Syllabus: Pythagoras' theorem (Ch 2 review ER 2a)	**10 cm**	2	1 mark: $6^2 + 8^2 = 100$ (or $c^2 = 100$). 1 mark: c = 10 cm.
35	Trigonometry: finding a side Syllabus: Trigonometry	**5.6 cm**	2	1 mark: x = 8 × tan 35°. 1 mark: 5.6 cm. (A calculator in radians mode gives 3.8, in gradians mode 4.9: 1 mark for the working.)

General marking notes

- Give full marks for a correct answer, even with no working.
- Where a part uses an earlier answer, follow through: mark the method using the student's earlier answer.
- Do not take marks off for a missing $ sign or unit if the number is correct; note it for feedback.
- Rounding: give full marks for a correct value rounded to a different number of decimal places than asked (for example 18.85 instead of 18.8); note it for feedback.

Reporting

Each section is one content area, so each section total can be reported on its own.

Section	Content area	Where it comes from	Marks	Share of test
A	**Financial mathematics** Percentages, GST, profit, discount, wages, loans, interest	Chapter 1 review and booklet	15	27.3%
B	**Measurement** Units, perimeter, circles, area, surface area, volume	Chapter 2 review and booklet	15	27.3%
C	**Statistics and probability** Types of data, stem-and-leaf, mean, median, graphs, probability	Chapter 4 review and booklet	14	25.5%
D	**Other Year 10 topics** Index laws, algebra, equations, straight-line graphs, Pythagoras, trigonometry	Rest of the Year 10 syllabus	11	20.0%
	Chapter reviews and booklets (Sections A–C)	Chapters 1, 2 and 4	**44**	**80.0%**
	Rest of the Year 10 syllabus (Section D)	Indices, algebraic techniques, equations, linear relationships, Pythagoras' theorem, trigonometry	**11**	**20.0%**

CLASS RECORD

Student	A / 15	B / 15	C / 14	D / 11	Total / 55	%
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						
16						
Class average						

SECTION A

Financial mathematics

Questions **1–11** · **15 marks** · Suggested time **20 minutes**

____ / 15
Marker

Questions 1–4 are **multiple choice**. Circle the letter of the correct answer, like this: (B)

1 What is 28% of $89? **1 mark**

A $28.00 **(B) $24.92** C $64.08 D $113.92

HELP **Example** 10% of $50:
0.10 × 50 = $5

2 A price tag says "**$55 (GST inc.)**". How much GST is included in the price? **1 mark**

(A) $5.00 B $5.50 C $50.00 D $60.50

HELP **Remember** GST in a price (GST inc.) = price ÷ 11

3 A shirt costs $56. It is **discounted by 26%**. What is the new price? **1 mark**

A $14.56 B $26.56 C $30.00 **(D) $41.44**

HELP **Remember** Find 26% of $56, then **take it off** the price.

4 Find the **simple interest** on $600 at 5% per year for 4 years. **1 mark**

A $12.00 B $30.00 **(C) $120.00** D $720.00

HELP **Formula** $I = P \times R \times N$
(5% → 0.05)

5 Increase $968 by 12%. Use a multiplier and your calculator. **1 mark**

Working

HELP

Multiplier Increase by 8% → × 1.08
Decrease by 8% → × 0.92

Example Increase $50 by 10%:
50 × 1.10 = $55

Answer $ **1 084.16**

6 A shop buys a cap for \$7.60. This is the **cost price**. The shop adds a **50% mark-up**. **2 marks**

a Find the selling price of the cap.

Working

Answer \$ **11.40**

b How much profit does the shop make?

Working

Answer \$ **3.80**

HELP
Remember Mark-up of 30% → × 1.3
Profit = selling price – cost price
Example Cost \$10, mark-up 20%: 10 × 1.2 = \$12, profit = \$2

7 Tom is paid \$15.78 per hour. **2 marks**

a Tom works 30 hours. How much is he paid?

Working

Answer \$ **473.40**

b Tom also works 4 hours of overtime at **time and a half**. How much is he paid for the overtime?

Working

Answer \$ **94.68**

HELP
Remember Pay = hours × hourly rate
Time and a half = 1.5 × hourly rate
Example \$20 per hour, 3 hours at time and a half: 3 × 1.5 × 20 = \$90

8 Pedro sells phones. He earns **4.5% commission** on his sales. His sales this week are \$790. How much commission does he earn? **1 mark**

Working

Answer \$ **35.55**

HELP
Remember 4.5% → 0.045
Commission = 0.045 × sales

9 Tranh works 36 hours at $28.89 per hour. **2 marks**

a Find Tranh's **gross pay** (total pay before anything is taken out).

Working

Answer $ **1 040.04**

b Tranh pays $142.59 in tax and $22.50 in super (superannuation). Find his **net pay** (the pay he takes home).

Working

Answer $ **874.95**

HELP

Remember Gross pay = hours × hourly rate
Net pay = gross pay − tax − super

10 Mia buys a phone on a payment plan. She borrows $1 200 and repays **$60 a month for 2 years**. How much **interest** does she pay? **1 mark**

Working

Answer $ **240.00**

HELP

Remember 2 years = 24 months
Total repaid = repayment × number of months
Interest = total repaid − amount borrowed

Example Borrow $500, repay $50 a month for 1 year:
50 × 12 = $600
Interest = 600 − 500 = $100

11 $5 000 is invested for 3 years. It earns 4% **compound interest** each year. **2 marks**

a Find the value of the investment after 3 years.

Working

Answer $ **5 624.32**

b How much interest is earned?

Working

Answer $ **624.32**

HELP

Formula $A = P(1 + R)^n$
P = amount invested, R = rate as a decimal, n = number of years

Remember 5% → 1 + 0.05 = 1.05

Example $2 000 at 5% for 2 years:
$A = 2000 \times 1.05^2 = \$2\,205$
Interest = 2205 − 2000 = $205

Calculator Type 2000 × 1.05 ^ 2 =

SECTION B

Measurement

Questions **12–21** · **15 marks** · Suggested time **20 minutes**

____ / 15
Marker

Questions 12–15 are **multiple choice**. Circle the letter of the correct answer, like this: (B)

12 How many metres are in 1 kilometre? **1 mark**

(A) 10 (B) 100 **(C) 1 000** (D) 10 000

13 What is the **area** of a rectangle 3 cm long and 4 cm wide? **1 mark**

(A) 7 cm^2 (B) 12 cm (C) 14 cm **(D) 12 cm^2**

14 What is the **volume** of a cube with sides 3 cm long? **1 mark**

(A) 9 cm^3 **(B) 27 cm^3** (C) 27 cm^2 (D) 54 cm^3

HELP **Formula** Volume of a cube = side × side × side

15 Which of these is 232 000 written in **scientific notation**? **1 mark**

(A) 2.32×10^5 (B) 23.2×10^4

(C) 2.32×10^{-5} (D) 232×10^3

HELP **Example** $2\,350\,000 = 2.35 \times 10^6$ (one digit before the decimal point)

16 Convert each measurement. **2 marks**

a Change 5.3 km into metres.

Answer **5 300** m

b Change 2.5 L into millilitres.

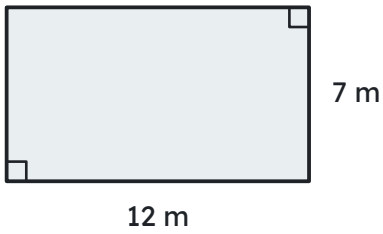
Answer **2 500** mL

HELP

Remember km → m: × 1000
L → mL: × 1000

17 A farmer puts a fence all the way around this rectangular paddock. How many metres of fence are needed? **1 mark**

HELP
Remember Perimeter = add up the lengths of **all** the sides.



Working

Answer 38 m

18 This circle has a radius of 3 m. Give your answers to 1 decimal place. **2 marks**

a Find the **circumference**.

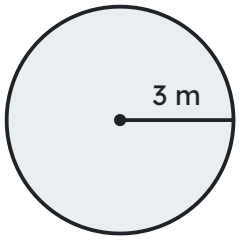
Working

Answer 18.8 m

HELP
Formula $C = 2 \times \pi \times r$ (circumference)
$A = \pi \times r \times r$ (area)
Example Radius 5 cm:
$C = 2 \times \pi \times 5 = 31.4$ cm
$A = \pi \times 5 \times 5 = 78.5$ cm^2

b Find the **area**.

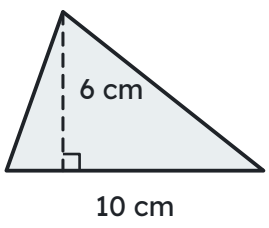
Working



Answer 28.3 m^2

19 Find the area of each shape. **2 marks**

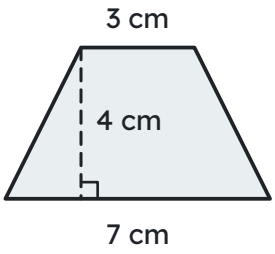
a Triangle



Working

Answer 30 cm^2

b Trapezium

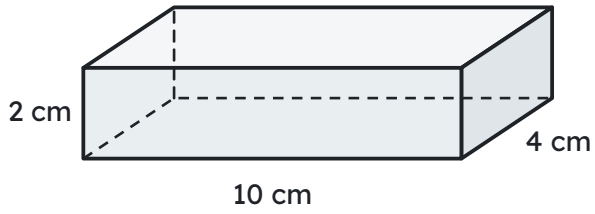


Working

Answer 20 cm^2

HELP
Formula Triangle: $A = ½ \times$ base $\times$ height
Trapezium: $A = ½ \times$ (top + bottom) $\times$ height
(top and bottom are the parallel sides)
Example Triangle, base 8 cm, height 5 cm:
$½ \times 8 \times 5 = 20$ cm^2

20 Find the **surface area** of this box. Complete the table. The first row is done for you. **2 marks**



HELP

Remember A box has 6 faces, in 3 matching pairs. Surface area = add the areas of all 6 faces.

Faces	Area of one face	Number of faces	Area
Top and bottom	$10 \times 4 = 40$ cm^2	2	80 cm^2
Front and back	**$10 \times 2 = 20$ cm^2**	2	**40 cm^2**
The two ends	**$4 \times 2 = 8$ cm^2**	2	**16 cm^2**
Surface area (add the last column)			**136 cm^2**

21 This cylinder has a radius of 2 cm and a height of 6 cm. **2 marks**

a Find the **volume**, to 1 decimal place.

Working

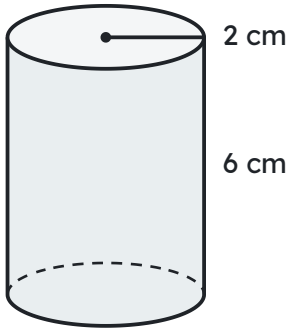
HELP

Formula Volume of a cylinder: $V = \pi \times r \times r \times h$

Example $r = 3$ cm, $h = 10$ cm: $V = \pi \times 3 \times 3 \times 10 = 282.7$ cm^3

Remember 1 cm^3 holds 1 mL

Answer **75.4** cm^3



b How many millilitres (mL) of water can it hold?

Working

Answer **75.4** mL

SECTION C

Statistics and probability

Questions **22–29** · **14 marks** · Suggested time **20 minutes**

____ / 14
Marker

Questions 22–25 are **multiple choice**. Circle the letter of the correct answer, like this: (B)

22 A survey asks: **"What is your favourite sport to play?"** What type of data does this give? **1 mark**

(A) numerical and discrete
(B) numerical and continuous
(C) categorical and ordinal
(D) categorical and nominal

HELP
Remember Categorical = words or groups. Numerical = numbers.
Nominal = no order (eye colour, type of pet).
Ordinal = has an order (small, medium, large).

Use this **stem-and-leaf plot** for Questions **23** and **24**.
It shows the scores of 11 players in a game.

Stem	Leaf
2	4 9
3	1 1 7 8
4	2 4 6
5	0 4

4 | 2 means 42

23 What is the **lowest** score? **1 mark**

(A) 4 (B) 20 (C) 24 (D) 54

24 What is the **mode** (the most common score)? **1 mark**

(A) 3 (B) 31 (C) 38 (D) 42

25 A normal 6-sided dice is rolled once. What is the probability of rolling an **even** number? **1 mark**

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$

HELP **Formula** Probability = number of ways it can happen ÷ total number of ways

26 These are the goals a netball player scored in 7 games: **3 marks**

8, 3, 13, 1, 5, 3, 9

Write the numbers in order first: 1 3 3 5 8 9 13

a Find the **mean**.

Working

Answer 6

b Find the **median**. Answer 5

c Find the **range**. Answer 12

HELP

Mean	Add up the numbers, then divide by how many numbers there are
Median	The middle number, when they are in order
Range	Biggest – smallest

27 One letter is chosen at random from the word **2 marks**

I N T E R E S T

a Find the probability that the letter is an **E**.

Answer $\frac{2}{8}$ or $\frac{1}{4}$

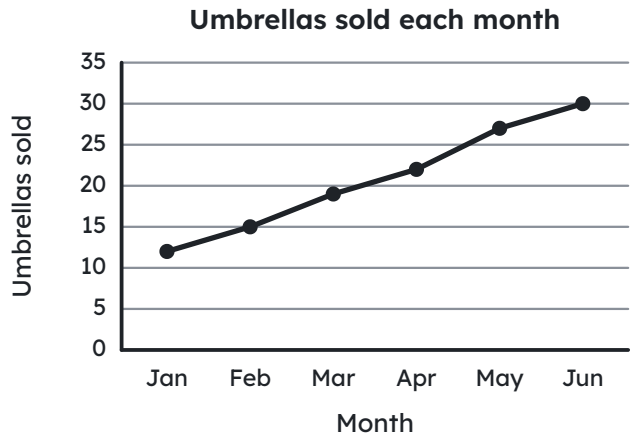
b Find the probability that the letter is a **vowel** (A, E, I, O or U).

Answer $\frac{3}{8}$

HELP

Formula	Probability = number of ways it can happen ÷ total number of ways
Example	A letter from **BOOK**: P(O) = $\frac{2}{4} = \frac{1}{2}$

28 This **time-series graph** shows how many umbrellas a shop sold each month. **1 mark**



HELP

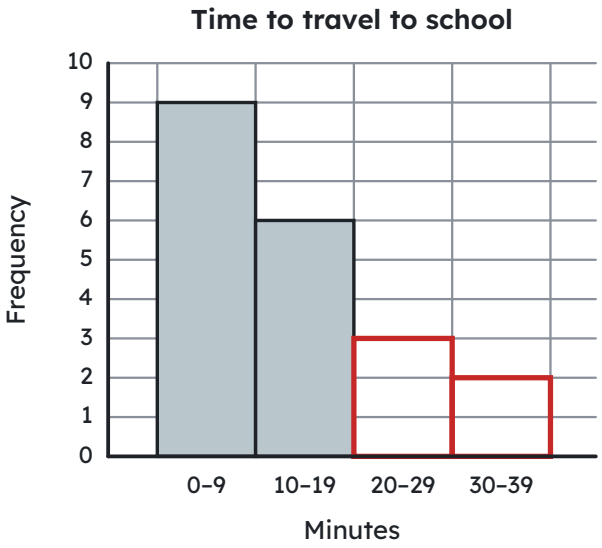
Remember	Upward trend: the line goes up over time. Downward trend: the line goes down over time.

Describe the **trend**. Circle one.

upward trend | **downward trend** | **no trend**

29 Some students recorded how many minutes it takes them to travel to school. **4 marks**

Minutes	Frequency
0–9	9
10–19	6
20–29	3
30–39	2
Total	**20**



a Fill in the **total** at the bottom of the table.

Total = 9 + 6 + 3 + 2 = 20

b The graph has 2 columns drawn. Use the table to **draw the columns** for 20–29 and 30–39 on the graph.

Columns of height 3 and 2, touching, shown on the graph.

c What **percentage** of the students took 0–9 minutes?

Working

HELP

Formula Percentage = frequency ÷ total × 100

Shapes



Answer 45 %

d Is the data **symmetrical** or **skewed**? Circle one.

symmetrical **skewed**

SECTION D

Other Year 10 topics

Questions **30–35** · **11 marks** · Suggested time **15 minutes**

____ / 11
Marker

Question 30 is **multiple choice**. Circle the letter of the correct answer, like this: (B)

30 Simplify $a^3 \times a^4$ — **1 mark**

(A) a^7 (B) a^{12} (C) $2a^7$ (D) $7a$

HELP **Remember** When you multiply, **add** the powers. $m^2 \times m^5 = m^7$

31 If $a = 4$, find the value of $5a - 2$. — **1 mark**

Working

HELP
Remember $5a$ means $5 \times a$
Example If $a = 3$:
$2a + 1 = 2 \times 3 + 1 = 7$

Answer 18

32 Solve each equation. — **2 marks**

a $2x + 3 = 13$

Working

Answer $x =$ 5

HELP
Remember Do the **opposite** to both sides.
Example $3x + 1 = 13$
Take away 1: $3x = 12$
Divide by 3: $x = 4$
Example $\frac{x+1}{2} = 4$
Times by 2: $x + 1 = 8$
Take away 1: $x = 7$

b $\frac{x+2}{3} = 4$

Working

Answer $x =$ 10

33 The rule for a straight line is $y = 2x + 1$. **3 marks**

a Use the rule to find y when $x = 0$, 1, 2 and 3. Write each answer in the table. 1 mark

x	0	1	2	3
y	**1**	**3**	**5**	**7**

1 mark: at least three of the four values correct.

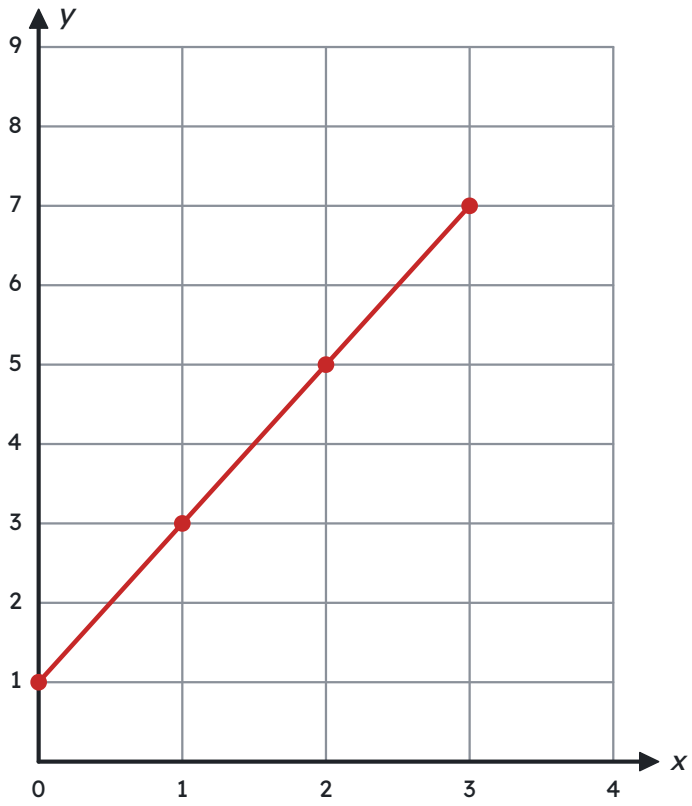
b Plot the points on the grid and join them with a straight line. 1 mark

Points plotted correctly and joined with a ruled straight line (follow-through from their table).

c Write the **gradient** and the ***y*-intercept** of $y = 2x + 1$. 1 mark

Gradient = **2**

y-intercept = **1**



HELP

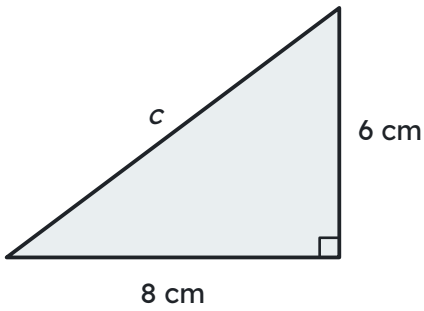
Example For $y = 3x + 2$, when $x = 4$: $y = 3 \times 4 + 2 = 14$

Plot For $x = 1$ and $y = 3$: go across to 1, then up to 3. Put a dot. Join the dots with a ruler.

$y = mx + c$ m is the gradient (how steep the line is). c is the y-intercept (where the line crosses the y-axis).

34 Find the length of the **hypotenuse**, *c*. **2 marks**

Working



HELP

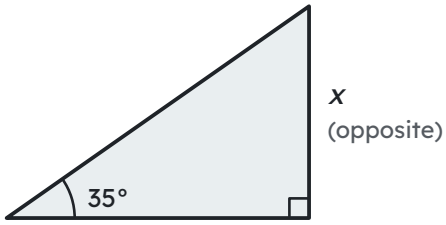
Formula $c^2 = a^2 + b^2$ (*c* is the longest side)

Example $a = 3$, $b = 4$:
$c^2 = 9 + 16 = 25$
$c = \sqrt{25} = 5$

Answer $c =$ **10** cm

35 Find the length of side *x*, to 1 decimal place. **2 marks**

Working



HELP

Formula tan (angle) = opposite ÷ adjacent
so opposite = adjacent × tan (angle)

Example Angle 20°, adjacent 5 cm:
$x = 5 \times \tan 20° = 1.8$ cm

Calculator Check your calculator shows **D** or **DEG** (degrees).

Answer $x =$ **5.6** cm

End of the test

- Did you answer **every** question? Have a go at any you missed.
- Did you write the **units** ($, cm, m^2, mL) and show your **working**?
- Use the space below if you need more room. Write the question number next to your working.

Extra working space